

ROSHAN DHARAN

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EDUCATION

University of Southern California

Master of Science in Computer Science

Aug 2025 – Present

Los Angeles, CA

REVA University

Bachelor of Technology in Computer Science and Engineering, GPA: 3.97/4

Jul 2019 – Jul 2023

Bengaluru, India

- **Coursework:** Data Structures and Algorithms, Object-Oriented Programming (Python/Java/C++), Advanced DBMS, Operating Systems, Machine Learning, Artificial Intelligence, Cloud Computing, Big Data, Data Science with R

TECHNICAL SKILLS

Languages: Python, Java, C++, JavaScript, TypeScript, SQL (PostgreSQL, CCL), HTML/CSS, MATLAB

Backend and Frameworks: Spring Boot, FastAPI, REST API Design, Node.js, Distributed Systems, Mockito

DevOps and Tools: Docker, Kubernetes, CI/CD (GitHub Actions), Git, Linux, Agile/Scrum, JIRA

Cloud and Databases: AWS (S3, EC2, Lambda), GCP, OCI, Snowflake, PostgreSQL, MongoDB, Redis

ML and AI: TensorFlow, Keras, PyTorch, Sklearn, XGBoost, LangGraph, LSTM, Transformers (BioClinicalBERT), NLP

EXPERIENCE

Keck Medicine of USC

Software Engineer Intern

Nov 2025 – Present

Los Angeles, CA

- Serve as Technical Lead for a 6-person Tiger Team, architecting a multi-layer predictive no-show system — a Snowflake data platform, two versioned XGBoost models, and a Keck Care API integration — surfacing real-time risk scores across tens of thousands of monthly appointments in Cerner RevCycle.
- Fine-tuned a transformer-based summarization model (LoRA on a T5/BART backbone) in Keck's AI sandbox to auto-generate discharge summaries, then engineered a Python/LaTeX QA pipeline catching errors across 2,000+ responses, cutting clinician correction time by an estimated 30%.
- Shipped a backend service ingesting Cerner ADT data to track real-time bed capacity across units, layering a lightweight predictive classifier to flag capacity strain hours in advance, presented as a technical workshop at a Spring 2026 hackathon.

Oracle

Associate Software Engineer

Oct 2023 – Jul 2025

Bengaluru, India

- Architected scalable, concurrent backend pipelines (CCL, PL/SQL) on Oracle Health's Cerner Millennium platform, processing 600,000+ patient records daily to deliver financial reporting infrastructure informing \$200M in revenue decisions across 17 departments.
- Developed a 340B Extract service using CCL, PL/SQL, and MoveIT integration with Pharmacy Med Manager, reducing pharmacy drug spend by \$12M while reclaiming 114 hours of manual effort annually.
- Implemented a CLABSI Audit backend service integrating PowerChart and Mobile Vitals Collection APIs via CCL, automating real-time monitoring for 500 patients weekly and eliminating 15 hours of manual nursing checks weekly.
- Optimized MedAnalytics reporting infrastructure across 17 hospital departments using HealthIntent and PL/SQL, engineering backend ETL workflows that cut report generation time by 40% and improved system reliability.

Oracle

Software Engineering Intern

Jan 2023 – Jul 2023

Bengaluru, India

- Modernized 5 on-premise Power BI systems to Snowflake Cloud, rebuilding SQL data models into a unified schema and automating data entry via a concurrent 8-thread Python/Selenium engine, reducing dashboard load time by 35% and manual effort from 44 hours to under 3 minutes.

PROJECTS

Hospital AI Command Center — LangGraph, XGBoost, SHAP, ChromaDB, Claude, FastAPI

Mar 2026 – Jul 2026

- Deployed a real-time multi-agent system (LangGraph StateGraph, conditional routing) firing on every ADT event, running five XGBoost risk classifiers with SHAP explainability at 0.77–0.83 validation AUC.
- Built a Claude-powered intervention agent that synthesizes five risk scores and a ChromaDB RAG layer over 10 CMS/HEDIS documents into structured JSON action plans with named assignees, logged to PostgreSQL and broadcast live via SSE with Accept/Override/Escalate attribution.

RAG-Powered Codebase Q&A Agent — LangChain, ChromaDB, sentence-transformers

Feb 2026 – Apr 2026

- Created a RAG pipeline using language-aware chunking and MMR retrieval over locally-embedded vector search, ingesting any GitHub repo — Flask's 88 files into 573 chunks in 5.8 seconds — and answering cited codebase questions.
- Ran a 10-query retrieval eval, tuning MMR to raise hit-rate@k from 70% to 90%, then shipped git-diff-based incremental re-ingestion, a SQLite-backed repo registry, and size/timeout guards for production hardening.

HONORS & PUBLICATIONS

Oracle Pinnacle Award — Invented memory allocation method adopted org-wide.

Jun 2025

"AI-Powered Battery Management System," peer-reviewed publication on EV battery SoC/SoH prediction — *International*

2023

Journal of Computational Learning & Intelligence